Sampling Distributions

Code

STAT 4714

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0.1 A Modern Perspective

- **Repeated sampling** is the foundation of inference
- Three levels: **Population → Sample → Sampling Distribution**
- Sampling and simulation can be used to motivate Ch 8-10 (i.e., hypothesis testing, confidence intervals, etc.)

0.2 Ex: Why not trust a single sample?

An engineer measures the voltage of a new circuit board three times:

- 5.12 V, 4.95 V, 5.08 V

Which is the "true" voltage? What happens if we **repeat the sampling process** many times?

0.3 The Three Levels of Variation

1. **Population** — full set of individuals/measurements with natural variability among units in population
2. **Sample** — what we actually measure (e.g., $n = 3$) with variation due to randomness
3. **Sampling distribution** — how a statistic (i.e., sample mean) varies across many hypothetical samples; quantifies randomness by showing how our statistic(s) fluctuates around the true population parameter

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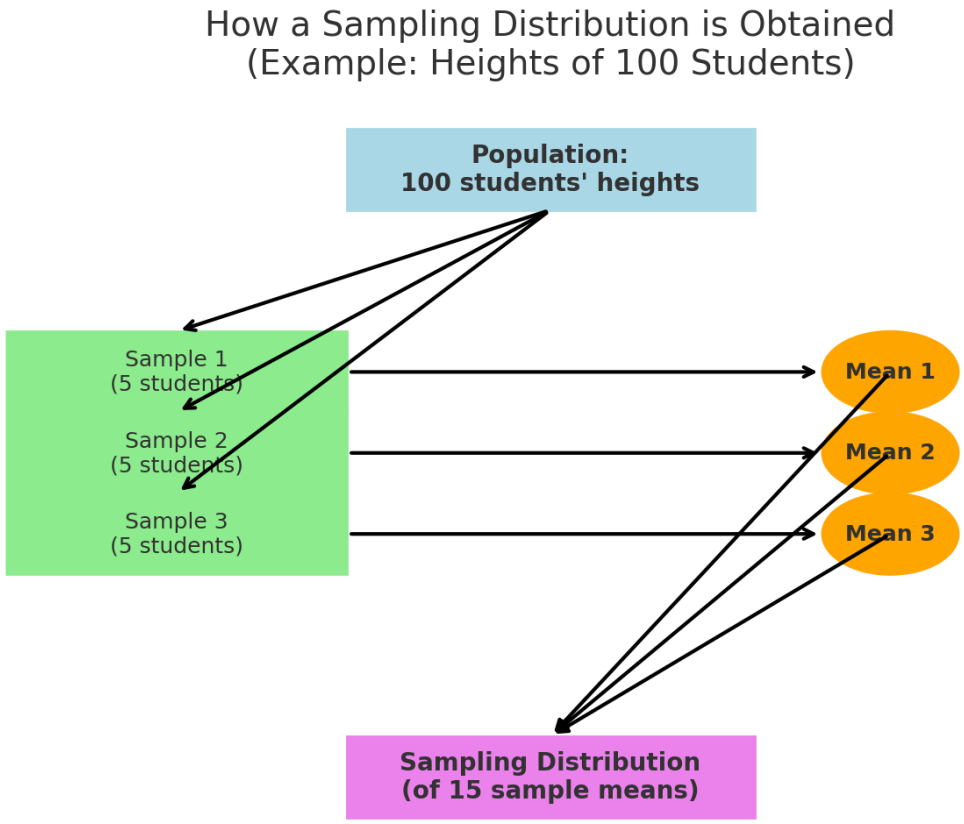
0.4 Ex: Height in Our Classroom

Consider the average height in our classroom of $n = 100$ students. Suppose that the true average height in this room is 5'6". While there is a truth, there is variability among everyone in this room (variability in population). Now, I ask 5 students to give me their height and compute the sample mean (randomness in sample). Is this sample average a good idea?

- Now, consider that I sample 5 students over and over, computing the sample mean each time. If I do this 15 times, I have 15 sample means. So, our **sampling distribution** is the distribution of sample means.

0.5 Visualizing the Sampling Distribution



0.6 TL;DR

- **Population**: broad variability of individual measurements
- **Sample**: one draw; could be atypical
- **Sampling distribution of the mean**: captures the uncertainty of the **estimator** and should better reflect the truth than a single sample mean

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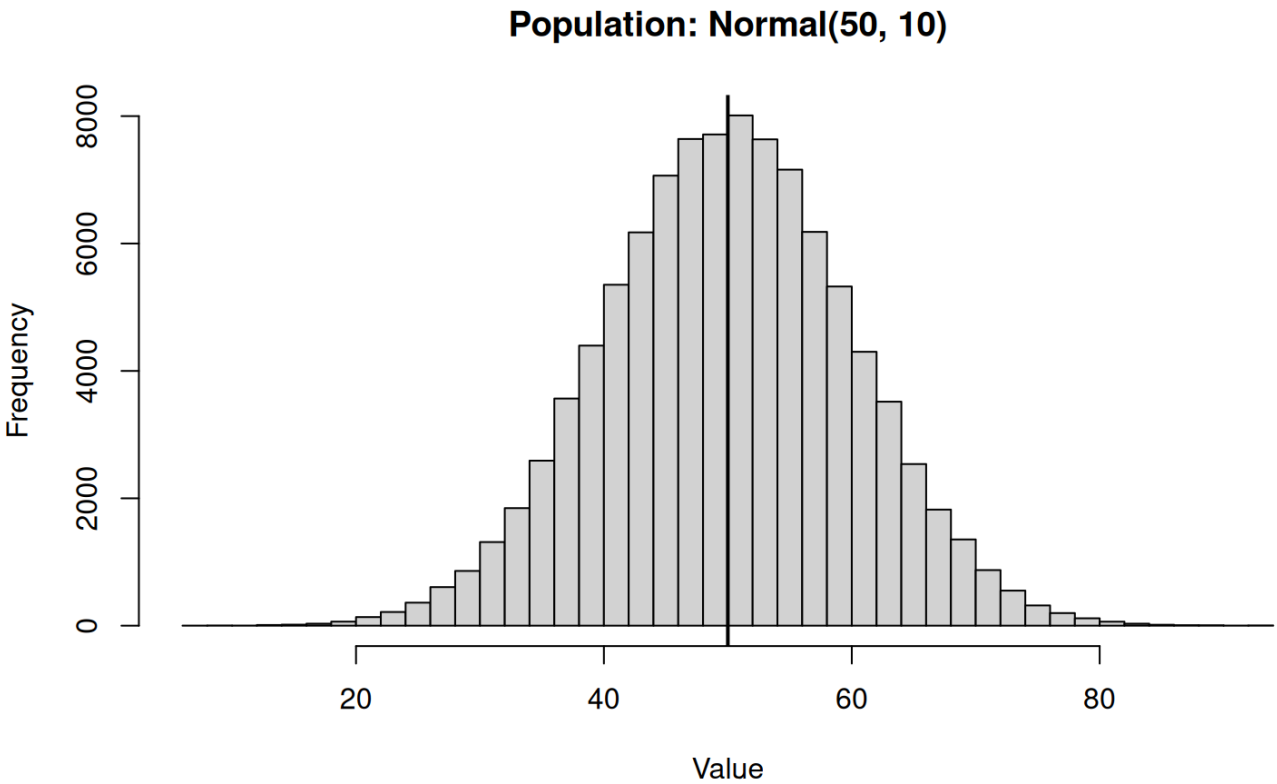
1.1 Sim #1: Normal Population

▼ Code

```
pop <- rnorm(100000, mean = 50, sd = 10)
hist(pop, breaks = 60, main = "Population: Normal(50, 10)", xlab = "Value")
abline(v = 50, lwd = 2)
```



Synthetic population: Normal(μ=50, σ=10).

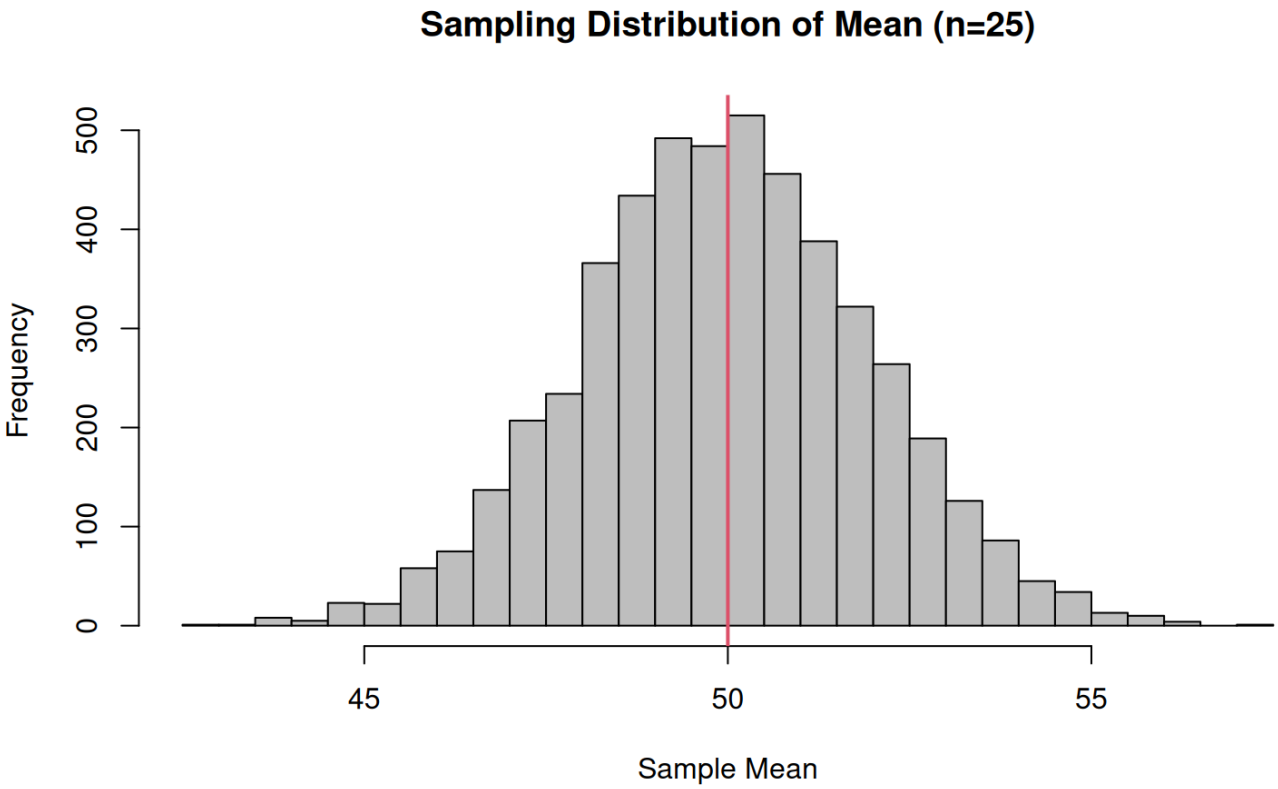
1.2 From One Sample to Many

- A **sample** of size *n* gives **one** sample mean $\bar{X}$
- If we repeat the sampling, we can get a **distribution** of $\bar{X}$ (the **sampling distribution**)

1.3 Sim #1a: Sampling Distribution for $\bar{X}$

▼ Code

```
n <- 25
n_sims <- 5000
means_n25 <- replicate(n_sims, mean(sample(pop, n)))
hist(means_n25, breaks = 40, col = "grey", main = "Sampling Distribution of Mean (n=25)",
abline(v = 50, col = 2, lwd = 2)
```



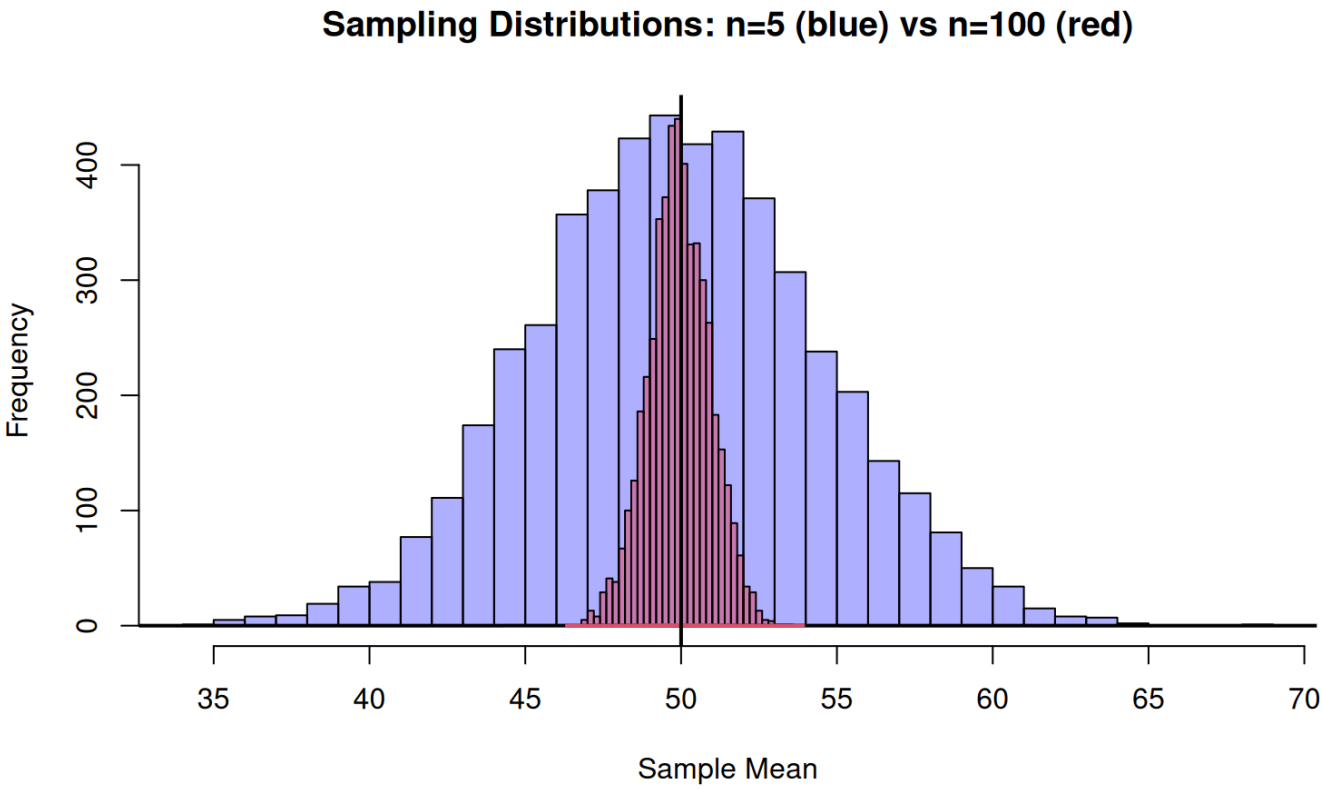
Sampling distribution of the mean for n=25.

1.4 Building Intuition

If we increase sample size to **n = 100**, what changes?

- a. Sampling distribution gets **narrower**
- b. Sampling distribution gets **wider**
- c. **No change**
- d. Center **shifts** away from the true mean

1.5 Simulation #1b: Small vs Large *n*



n=5 vs n=100: sampling distributions.

Conclusion: Larger n → **smaller standard error** → **narrower** sampling distribution.

1.6 Standard Error

- **Standard error (SE)** is the standard deviation of a sampling distribution. It tells us how much a statistic (e.g., sample mean) varies if we repeatedly take random samples from the same population.
- Population standard deviation: σ
- The standard error for the sample mean is

$$\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}$$

- Side quest: if σ is unknown, we can estimate $\mathrm{SE}(\bar{X})$ (i.e., **t tests**)

1.7 Why Sampling Distributions?

- They let us quantify uncertainty and know how trustworthy our statistic is.
- They connect samples to populations and allow us to generalize back to the population.
- They enable and **motivate** inference, such as confidence intervals, t-tests, and F-tests.
- **They are the bridge between sample data and population conclusions.**

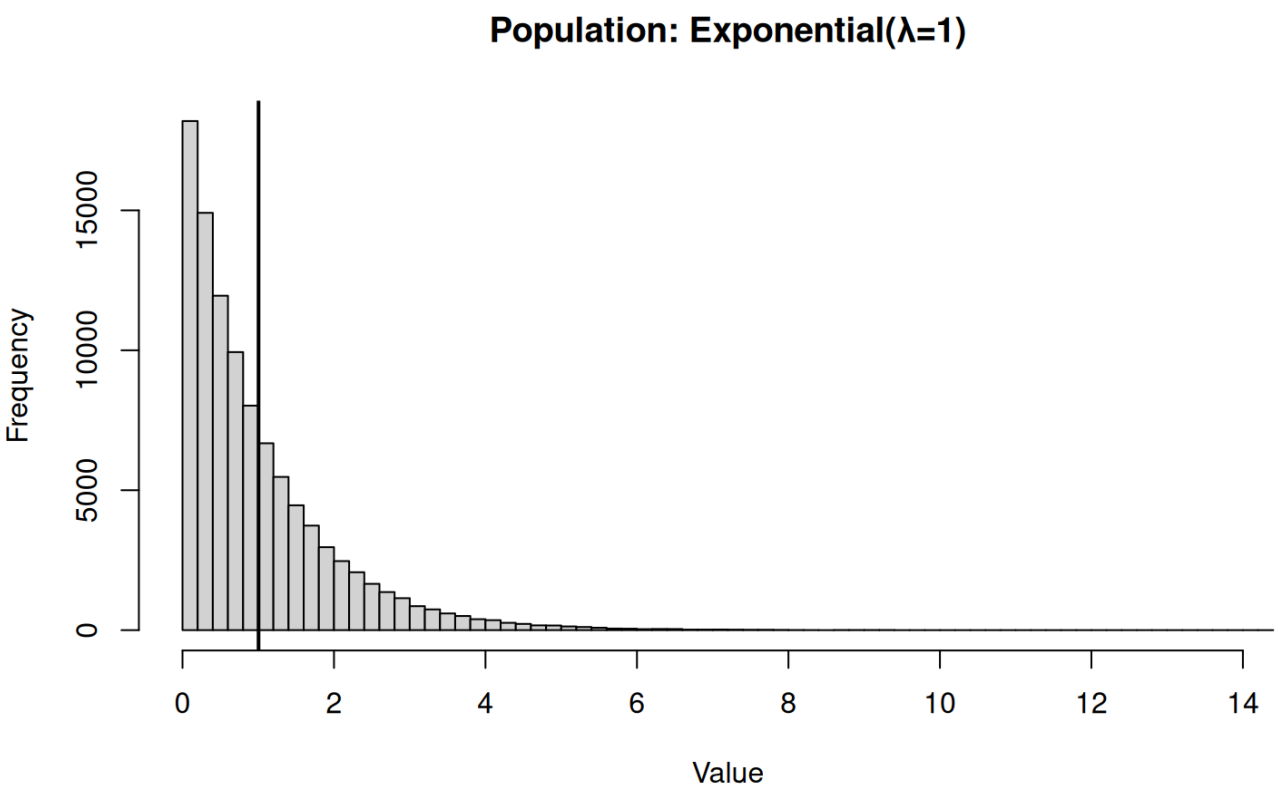
2.1 Non-Normal Populations Happen

- Real data often **aren't normal** (skewed, heavy-tailed).
- What happens to the sampling distribution of the **sample mean**?
- **Central Limit Theorem (CLT)**: for large *n*, the sample mean is **approximately normal**.
  - *Note*: Our next lecture will be dedicated to the CLT.

2.2 Sim #2: Exponential Population

▼ Code

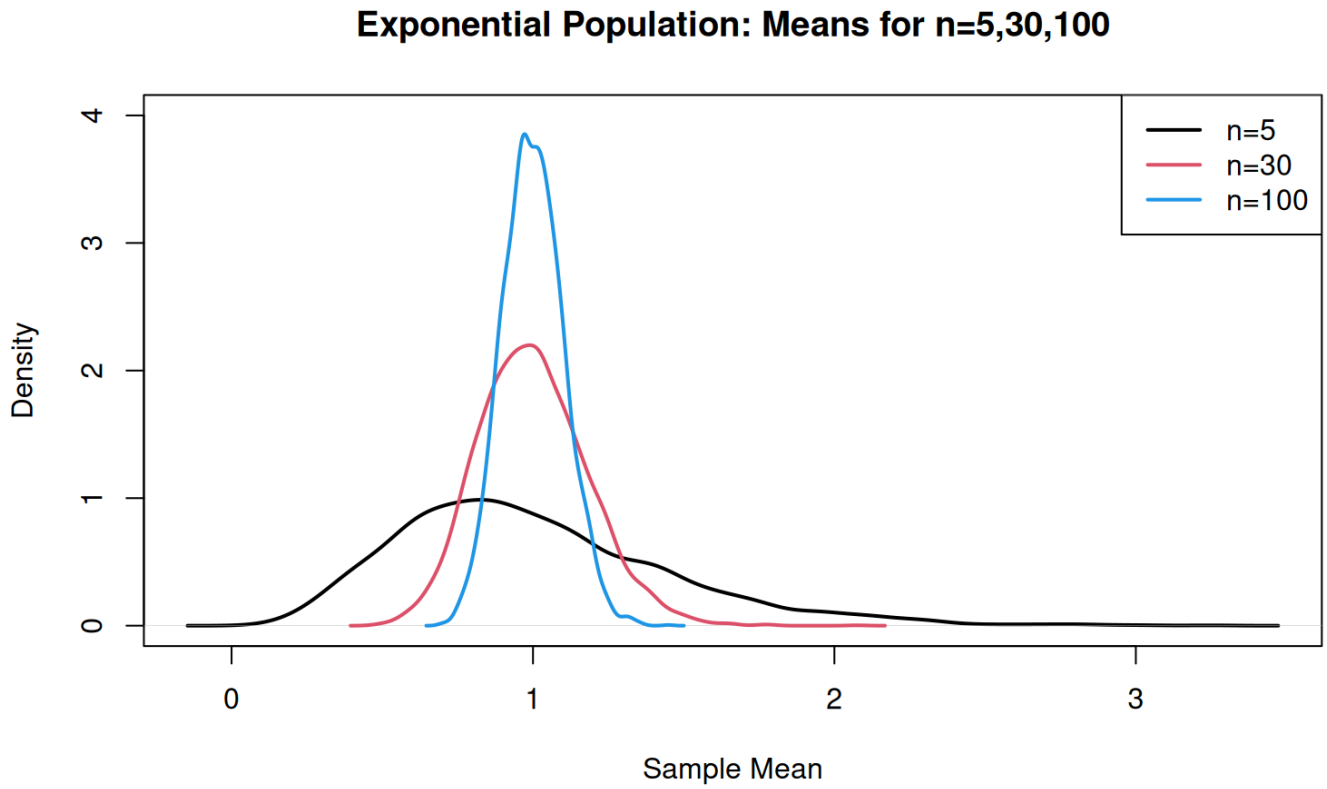
```
pop_exp <- rexp(100000, rate = 1)
hist(pop_exp, breaks = 80, main = "Population: Exponential(λ=1)", xlab = "Value")
abline(v = mean(pop_exp), lwd = 2)
```



Population: Exponential(1) — skewed.

2.3 Sampling Means from a Skewed Population

▶ Code



Means from Exponential(1): n=5 vs 30 vs 100.

3.1 Beyond Means: Proportions in Reliability

- Many ECE questions are **defect/success rates** (Bernoulli)
- Statistic: **sample proportion** $\hat{p}$
- The **sampling distribution** of $\hat{p}$ across repeated samples quantifies the uncertainty of the reliability

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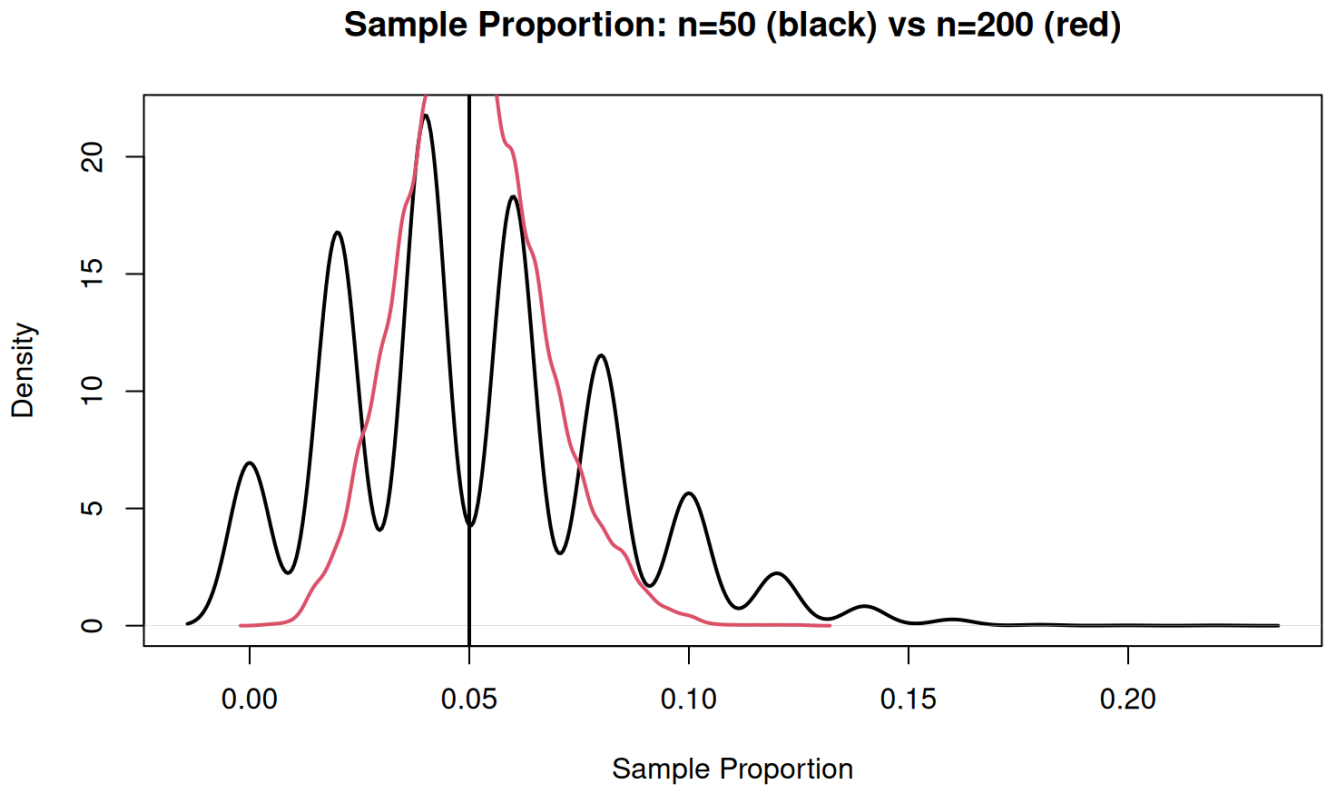
3.2 Sim #3: Proportion Defective

▶ Code



Sample proportion with p=0.05: n=50 vs n=200.

Conclusion: Larger n → smaller **SE** for $\hat{p}$ → more precise reliability estimates.

3.3 A Modern Perspective: Moving Forward

- Instead of formulas first, **simulate** the appropriate sampling distribution under a hypothesized model
- Extends to **bootstrap CIs**, **permutation tests**, **randomization tests**
- Today, we build intuition. Later, we'll see classical formulas (**t**, **F**) as **special cases**.

3.4 Common Misconceptions

- "A single sample mean is the population mean." → No, it's an estimate with uncertainty.
- "More data always removes bias." → Precision increases with n, bias is **systematic**.
- "Normality of data is required for inference." → If this were true, we would only be able to model a limited amount of processes!

3.5 Practical Tips

- When unsure about assumptions, **simulate** to build intuition before applying a test.

- Always report an estimate *and* a measure of uncertainty (SE or CI).
- Keep data, code, and seeds for reproducibility.
- Visualize the **sampling distribution** alongside numeric summaries.

4.1 Today's Key Ideas

- The **sampling distribution** describes how a statistic varies under repeated sampling.
- Larger samples → **smaller SE** → more precise estimates.
- This perspective unifies **confidence intervals** and **hypothesis tests**.
- Next: **Central Limit Theorem** & formalizing **standard error**.

4.2 Breakout Work

1. In your own words, what is a **sampling distribution**? Give a concrete example.
2. Name **two** engineering decisions where the variability of a **sample mean** is crucial.
3. If the population is **highly skewed**, what happens to the sampling distribution of the **mean** as n grows? Why?
4. How would you **estimate uncertainty** if you didn't trust normality assumptions?